

Creativity: what does it mean?

NPTEL

*MOOC
Soft skills
development*

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Overview

- Body language
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- Slides and PPTs
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Quiz: How creative are you?

- How creative are you? (Mindtools)

<https://www.mindtools.com/pages/article/creativity-quiz.htm>

Please enter your self-assessment scores in the Discussion Forum Link provided so that we can know what is the general level of communication skills of this class.

Defining creativity

- It is the ability to make or otherwise bring into existence something new, whether a new solution to a problem, a new method or device, or a new artistic object or form.

Encyclopedia Britannica

Creating novel useful products...

Wikipedia

- Creativity – as conventionally understood
- Scientist, Artist
 - Combination of creativity, skills, intelligence, natural abilities
- But the ability to generate new ideas is relatively easy

Models

Graham Wallas: *Art of Thought*, 1926

- (i) *preparation* (preparatory work on a problem that focuses the individual's mind on the problem and explores the problem's dimensions)
- (ii) *incubation* (where the problem is internalized into the unconscious mind and nothing appears externally to be happening)
- (iii) *intimation* (the creative person gets a "feeling" that a solution is on its way)
- (iv) *illumination* or insight (where the creative idea bursts forth from its preconscious processing into conscious awareness)
- (v) *verification* (where the idea is consciously verified, elaborated, and then applied).

James C. Kaufman and Beghetto:

- personally *meaningful interpretations of experiences, actions and insights*
- everyday problem solving and creative expression
- Traits exhibited by people who are professionally or vocationally creative though not necessarily eminent
- creativity considered great in the given field

Little C & Big C

- J. P. Guilford: *The Nature of Human Intelligence*, 1967.
- Convergent thinking
- Divergent thinking / flexible thinking

Hellis and Sun, 2010: Explicit-Implicit Interaction

- Explicit and implicit processing of information and their relation
- Computationally oriented

IQ vs. Creativity

- De Bono's critique: *Intelligence is not enough*
- *Emotional Intelligence*, Daniel Goleman
- Creativity tests:
- Plot test, uncommon word association test, figure, remote association, etc (Guilford and others)

Important features highlighted...

- **Fluency** – The total number of interpretable, meaningful and relevant ideas generated in response to the stimulus.
- **Originality** – The statistical rarity of the responses among the test subjects.
- **Elaboration** – The amount of detail in the responses.
- Visual, auditory, associative, etc...
- Eg: Cricket – googly, helicopter shot

Brain and creativity

- Brain activity during creative behavior:
- creative innovation might require coactivation and communication between regions of the brain that ordinarily are not strongly connected.
- Highly creative people:
- they have a high level of specialized knowledge,
- they are capable of divergent thinking mediated by the frontal lobe
- they are able to modulate neurotransmitters in their brains

Kenneth M Heilman, MD, Stephen E. Nadeau, MD, and David Q. Beversdorf, MD. "Creative Innovation: Possible Brain Mechanisms"
Neurocase (2003)

Cultural differences in approaches

- Western creativity – more focus on the individual
- Eastern – collective cultures – social impact
- Africa and certain other places of the world do not have word equivalents to creativity, yet can possess creative arts
- Some countries associate creativity with the ability of coping with life
- Others may focus on just one aspect of creativity – say divergent thinking or problem solving...
- How do we, Indians, define creativity?

Still difficult to define...

- Intellect - intuition
- Conscious - unconscious
- Order - disorder
- Conventional -unconventional
- Left-brain - right-brain
- Originality, uniqueness, the different

Artistic vs. scientific

- Individual expression
- Divorce from social reality
- Expressive
- Disciplined
- Logical thinking

Problem-solving

- Two boys of different age are playing badminton. The older one is a more skilled player, and therefore it is predictable for the outcome of usual matches who will be the winner. After some time and several defeats the younger boy finally loses interest in playing, and the older boy faces a problem...
- What's the solution?

Solution by the older boy

- He proposes that they should try to keep the bird in play as long as possible - and thus changing from a game of competition to one of cooperation. They'd start with easy shots and make them harder as their success increases, counting the number of consecutive hits. The proposal is happily accepted and the game is on again.

Cognitive Psychology by Wikibook contributors, 2004-2006.<http://en.wikibooks.org/wiki/Cognitive_Psychology_and_Cognitive_Neuroscience>

Insight

- Solution to a problem based on past experience
- Solution which needs a radical new way of looking at things, a restructuring, nonobvious
- Ah! Light Bulb!
- Sometimes earlier knowledge or familiarity can make the problem for difficult – fixation

Why intelligence is not enough?

Edward De Bono

- Intelligence or mental ability?
- IQ tests
 - Ability – perception
 - Application – focus
- Intelligence is a potential
 - The car and driver analogy

- Intelligence

- Ego
- Ability to defend cogently
- Loss of opportunity to learn and explore
- Over confidence
- Analysis
- Argument
- Attack

To prove easily that someone is wrong – is this intelligent?

- Patterns and new patterns
 - Drop a random word to generate an whole range of new ideas!
 - Main roads and side roads
 - New possibilities

- Most creative ideas look logical when we look back at them? Why?
- But why do they sometimes look ridiculous at the beginning:
 - Large Computer – PC
 - Vaccination...

- New ways of thinking; new ways of teaching
 - De Bono; Sougato Mitra
 - Arguments
 - Search for truth
 - Logic
 - *Are they good enough for all situations...
convergent thinking?*

- Perception vs. logic
- David Perkins: Errors – perceptual or assumptive?
- Heuristic thinking
- Cognitive biases...
 - Bandwagon effect
- The mind is lazy

Changing perception is not a matter of intelligence but willingness...the possibilities are endless

- Design vs. analysis
 - Intelligence – truth
 - Design – possibilities and relevance

References and acknowledgement

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- Many of the slides are based on the webpage on creativity and problem solving at: <www.crinology.com>
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- *Cognitive Psychology* by Wikibook contributors, 2004-2006. [http://en.wikibooks.org/wiki/Cognitive Psychology and Cognitive Neuroscience](http://en.wikibooks.org/wiki/Cognitive_Psychology_and_Cognitive_Neuroscience)
- Helie S.; Sun R. (2010). "Incubation, insight, and creative problem solving: A unified theory and a connectionist model". *Psychological Review* **117**: 994–1024.

Creativity: A detailed exploration



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Objective

- What is creativity?
- Creative persons, and
- Creative process

- The challenge of 21 century is to have more innovation and creativity in every sphere of life—technology, health, education, agriculture, industry, environment, etc.
- Creative ideas from group and individuals are scarce.
For example students of business schools know how to develop models but they do not have knowledge of creative process and how to develop creative solution. Example- General food from USA, market plan to boost the plunging sale of “Sugar-free-cool-aid”.
- **Creativity is the production of new and original ideas. It encompasses novelty, originality, and usefulness. Creativity implemented is innovation.** It is popularly known as out-of-box thinking and mindmix procedure.

Are the two parts of the brain different?

- Why **out-of-box**? Inside the box we feel safe, everyone stays there with boundary of knowledge structure in mind. Out of box is invisible, it is accountable for divergent thinking.
- **The right hemisphere in brain processes divergent, non-linear, intuitive, visual-spatial, inductive and qualitative thoughts and responsible for creativity.** The left hemisphere processes convergent, linear, rational/analytical, verbal, deductive, quantitative thought.
- Right brain-idea generation; left brain-Idea evaluation and analysis; Both brains involved in idea protection and improvement. **Mindmix** for innovation.
- Socialization process and learning system condition us to use our left brain frequently, conform to the norm, write answer logically to fetch marks in exam, read a story and answer from the story and do not write similar new stories. Everybody has the right brain, recognize, respect, accept odd ideas, take steps to nurture and protect ideas

Creativity: Two tests

(1) Alternative Uses Test(Guilford, 1967)

Tell me as many unusual ways you could use a

1. Knife
2. Automobile Tire
3. Chair
4. Key
5. Pencil
6. Telephone book

Example

*Name all the uses for a **brick**:*

1. A paperweight
2. A doorstop
3. A mock coffin at a Barbie funeral
4. To throw threw a window
5. To use as a weapon
6. To hit my sister on the head with

Scoring is comprised of four components:

1. Originality - each response is compared to the total amount of responses from all of the people you gave the test to. Responses that were given by only 5% of your group are unusual (1 point), responses that were given by only 1% of your group are unique - 2 points). Total all the points. Higher scores indicate creativity*
 2. Fluency - total. Just add up all the responses. In this example it is 6.
 3. Flexibility - or different categories. In this case there are five different categories (weapon and hit sister are from the same general idea of weapon)
 4. Elaboration - amount of detail (for Example "a doorstop" = 0 whereas "a door stop to prevent a door slamming shut in a strong wind" = 2 (one for explanation of door slamming, two for further detail about the wind).
- *You might have noticed that the higher fluency the higher the originality (if you did "good for you!") This is a contamination problem and can be corrected by using a corrective calculation for originality (originality = originality/fluency).

(2) Consequences Test: Two forms with 5 items in each form(Guilford et al., 1980)

This is a test of your ability to think of a large number of ideas in connection with a new and unusual situation.

Below is a sample question:

What would be the results if people no longer needed or wanted sleep?

SAMPLE RESULTS:

1. Get more work done.
2. Alarm clocks not necessary.
3. No need for lullaby song books.
4. Sleeping pills no longer used.
5. _____

- What would be the result if none of us needed food any more in order to live?
- What would be the results if all the people in the world lost the ability to reproduce offspring?
- What would be the results if human life continued on earth without death?

Scoring:

Each judge was asked to rate the total pool of responses obtained for each problem (Runco & Mraz, 1992) on a 5-point scale ranging from 1 (low) to 5 (high) reflecting (1) quality, (2) originality, (3) time frame, (4) realism, (5) complexity, and (6) use of general principles. Two other divergent-thinking skills, (7) presence of positive consequences and (8) presence of negative consequences, were categorically rated yes (if present) or no (if not present).

What is the relation between IQ and Creativity? The relation between the two holds up to IQ 120.

Creative people:

- are divergent thinkers; newspaper → reading; tongue cleaner → cleaning tongue. Unusual ways how the items can be used.
- have cognitive complexity, terrorizing imagination, have preference for elaborate, intricate and complex stimuli and thinking
- are different from usual people, philosophical, mystiques. Capable, confident, clever, intelligent, egotistic, humorous, informal, individualistic, insightful, reflective, resourceful, sexy. They have wide range of interests, are independent, interested in abstract problems, have tolerance of ambiguity. Big five personality--Openness to experience (curiosity, flexibility, imaginativeness, openness to change, and unconventional ideas). They are more vulnerable to losing touch with reality that they are akin to madness.
- have exceptional mental and physical energy for pursuing their activity of interest. Inventors have a knack for persisting after failures as Thomas Edison tried countless substances for his light-bulb filament.
- are Intrinsically motivated; challenge of the task is more important than money, meeting deadline, secure job, etc.

Creative thinking process

- (1) Preparation: All education is a preparation for creativity. Einstein might not have discovered the theories of relativity and thermodynamics, had he not first learned the advanced physics and mathematics. Engineering problem-solving. Much trial and error like your term paper.
- (2) Incubation: banking or booking the ideas into the brain and thinking about the problem. Absence of overt activity and problem is solved unconsciously in mind. This is the stage where people go for reading light literature, engage in games, marry-making, go for sleeping. This continuance of associational activities, once stated, is referred to as perseverance (steady efforts to fulfill aims). So much engaged that do not get out, and some forget the problems.

- (3) Inspiration or illumination: This is the stage of sudden flash or insight. Creative ideas come suddenly –in dreaming, bathing, gardening, gossiping, etc. Archimedes law of buoyancy, during bathing, Kekule hexagonal shape of the Benzene[C_6H_6], a snake eating its own tail.
- (4) Verification: evaluate, test, and perhaps revise the ideas. Is the idea workable, useful, logical?

Everyone can be creative with proper training, zeal, and community and organizational encouragement and support. THANK YOU

Creativity: Activities

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Overview

- Key points
- Games
- Summary
- References

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Remember

- Separate generation of ideas from evaluation
- Assumptions need to be questioned
- Patterned thinking is best avoid - habitual
- Create new perspectives
- Negative thinking – the less the better
- Take careful risks

B Vangundy

Game 1: Combo Chatter

Also known as 'Semantic Intuition' Schaude, 1978

- To help participants develop as many creative ideas as possible
- Eg. Assume that an ice cream company wants a new ice cream flavor – *Kool Congo*. It may list ideas under, say, two categories:
 - Packaging
 - People who might have this ice-cream
- Make a list of five or six ideas under each category
- Link the ideas to come up with new concepts

Game 2: I like it I like that

- To help participants develop as many creative ideas as possible
- Problem: How to get more students interested in learning
- This problem is like
 - Making the dog interested in playing with you
 - Asking a child to learn a new game
 - Asking an old person to learn use of computers...

Some more methods

- Trial and error
- Means-end analysis
- Analogies
- Schema
- Divergent thinking

Analogies

- One uses an analogy when one says, “this is like that” (in some respects but not in others). For example: a plane is like an bird – they both fly, they both have wings, they can both travel for a long way without landing, and both can sense where they are going; but they are not similar in that they have different means of propulsion, are made of different things, etc.

- Analogies can be used creatively. For instance, biological analogies are often used to solve mechanical problems.

- “Identify what it is you want ideas for, and try to find a core verb phrase that captures the essential functional nature of what you are looking for - e.g.: 'How to make X'. 'How to prevent Y', 'How to speed up Z', 'How to become better at A'.”
- “For each verb phrase generate a list of items (people, situations, objects, processes, actions, places, etc.) that is 'like' it in some way - e.g. analogies to 'making X' (having a baby, making a pudding, the Genesis creation story, a robot car factory, ...etc.).”

- Identify analogies that look interesting - ideally from different domains.
- Describe the analogue:
Actions – how it works, what it does, what effects it has, how it is used.
Passive dimensions: size, position, etc.
- Can you relate these to find ideas relevant to your problem?
- Does the comparison be used directly?
- Do the differences suggest other possible ways of exploring your problem?

Attribute listing

- Attribute listing is a technique from the early 1930s.
- Take an existing system or product
- Break it down into small parts or subsets
- Identify various ways of getting these parts
- Can these be recombined to form new products or systems?

- One can identify products one wishes to improve.
- List its attributes. Say a pen:
 - Material, Shape, Target market, Colours, Textures, etc.
- Choose, say, 7-8 of these attributes that seem particularly interesting or important.
- Identify alternative ways to achieve each attribute.

Brainstorming

- Brainstorming and creative thinking often used interchangeably.
- Group-based generation of ideas
- Suspended

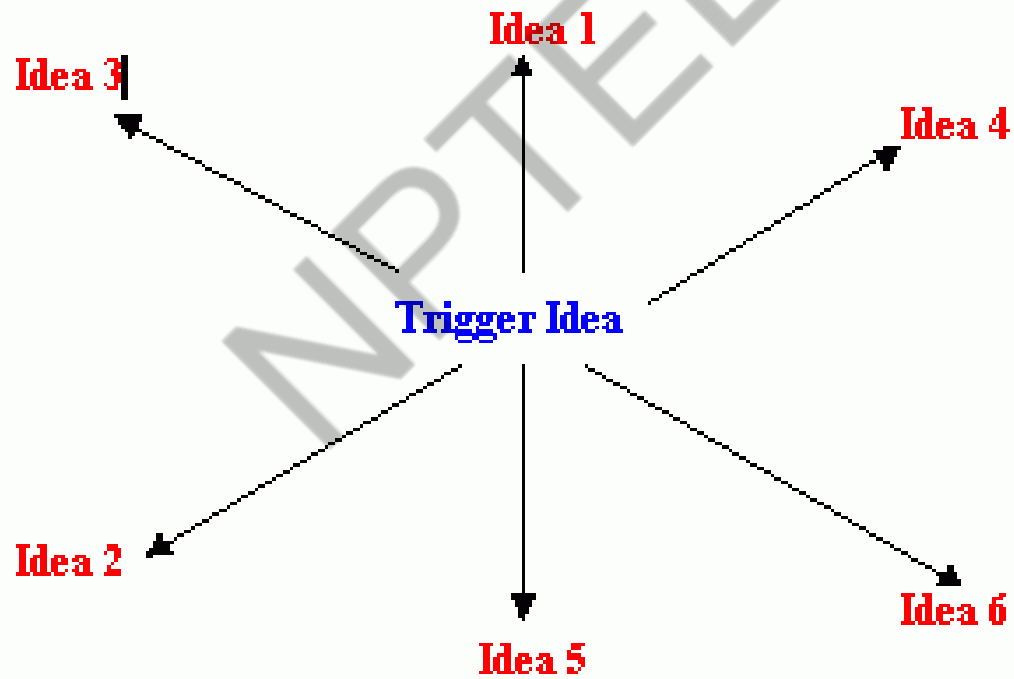
- A group of between 4-8 ideally
- Write the initial topic on a flipboard or whiteboard so that all can see it
- Clearly articulated problem can yield better solutions
- Has everyone understood the problem?
- Avoid criticising ideas / suspend judgement
- All ideas are equally valid

- Lots of ideas
- Don't criticize
- Listen to others, use their ideas to extend yours
- Avoid discussion that may stop the flow of ideas

- Someone should jot down the ideas
- Generate ideas
- Ideas should then be examined, classified, and the best options kept aside for further discussion

Free Association

- In serial association, start with a trigger, you record the flow of ideas that come to mind, each idea triggering the next, ultimately reaching a potentially useful one. It is like the stream of consciousness.
- Centred association: focus on one central trigger, around which ideas build up.



- **Suspend judgement.** Encourage ideas that you generally suspend to also come up.

- **Follow the fascinating and intriguing idea:**
- **They may not seem instantly apt, but may develop into meaningful ideas later.**

- **Use solution-oriented phrasing.**
- ‘Could we colour it red?’
- ‘In what ways might I make it ‘red?’
- ‘I wish it were ‘redder’
- ‘How might it help it if were redder’?’
- Makes the idea ‘red’ potentially a more useful one.

Six Thinking Hats

- Early in the 1980s [Dr. Edward de Bono](#) invented the Six Thinking Hats method. The method is a framework for thinking and can incorporate lateral or divergent thinking.
- The six hats stand for six modes of thinking. They are directions to think or encouragements to think in certain ways.

- It encourages Parallel Thinking
- It encourage full-spectrum thinking
- It separate ego/individual from performance



- **White Hat thinking**
- This covers facts, figures, information needs and gaps. Drop all arguments, look at facts and figures.



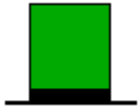
- **Red Hat thinking**
- Intuitions, feelings, emotions.
- Don't try to justify gut feelings, just state them.
- Full permission for coming forward with one's subjective feelings without the support of logic.



- **Black Hat thinking**
- Judgement, caution.
- Logical
- Use it to point out why something may not succeed



- **Yellow Hat thinking**
- Logical and positive.
- Look forward to something potential or look at something positive that has happened, and



- **Green Hat thinking**
- Creativity, alternatives, proposals, provocations, changes, etc.



- **Blue Hat thinking**
- Overview or process control
- Self-reflexive, looks at itself or the process of the different hat thinking.

References and acknowledgement

- Many of the slides are based on the webpage on creativity and problem solving at: <www.crinology.com>
- [Encyclopedia Britannica 2007, Deluxe Edition.](#)
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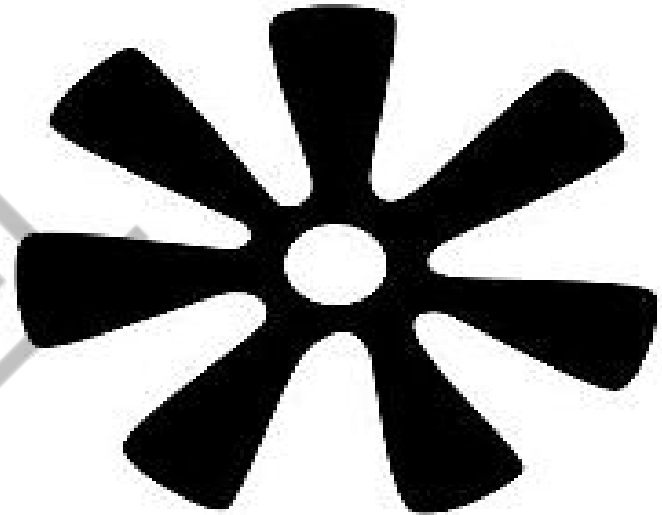
Creativity or Innovation at Workplace

Objectives:

Creativity and innovation

Innovation at workplace

Techniques for idea generation



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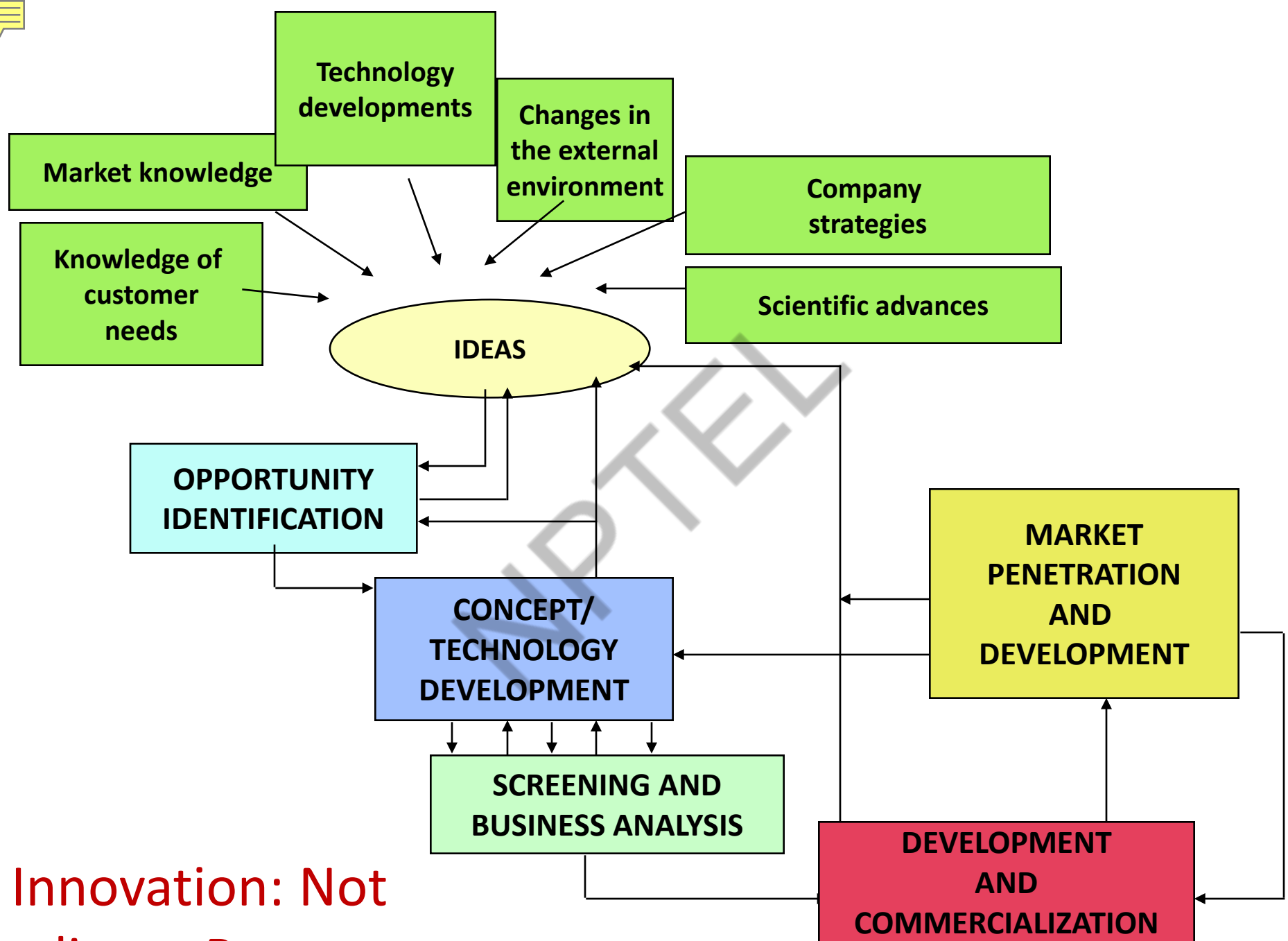
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- Workplace creativity is generally framed in the context of organizational products, services, processes, and procedures and focuses on the production of new and useful ideas (Amabile, 1996; Oldham & Cummings, 1996; Zhou, 1998).
- Implementation of creative ideas in workplace = innovation at workplace.
- Creative persons, groups, organizations (input)→ creative process in situation/environment (transformation)→ products, services, process (output)
- Innovation is the interplay between individual creativity and environmental creativity that is the driving force of innovation in any organization

New Challenge

- A company manufactures glassware. The last step of the process is to **wrap glasses individually in used newspapers and then place them in a specially designed box**. The box is then sealed. Management notices low productivity with this last step, observing that workers occasionally stop, to read the newspapers? What can management do to increase productivity?



**Innovation: Not
a linear Process**

Benefits of increased innovation ...

- Increased synergy from information sharing, joint initiatives, integrated policy, investment in R & D and benchmarking
 - Better efficiency and cheaper products and services which are more attractive than those offered by competitors
 - To foster innovation: Ensuring that organizational structures and processes optimize both creativity and innovation;
 - Developing the creative capacity of staff through training.
- How many organizations that exist today but that are not open to creativity and innovation and willing to change will still exist in 10 years time? **None!**
- There is need of continuous innovations for sustainability and profitability of organizations.

Understanding creativity and innovation involves understanding ...

- Exactly what is involved in the specific context of the organisation;
- What behaviour needs to be fostered;
- What supports and what inhibits creativity and innovation.

Some false assumptions ...

- Some people are creative while others are not;
- Creativity is something people have or don't have;
- Creativity is disruptive and counter-productive;
- Creativity is only relevant in certain departments (e.g., design, publicity, marketing)

Six barriers to organisational innovation ...

1. Lack of organizational slack
2. Bureaucracy
3. Structure
4. Poor Communication
5. The 'imported talent' syndrome
6. Financial aversion to risk taking

To overcome barriers ...

- Remove fear, encourage trust and sharing through better communication
- Make innovation part of everyone's performance review system
- Build in enough looseness for staff to explore new possibilities and collaborate with others both within and outside the system
- Train staff to scan the environment for new trends, technologies and changes in clients' mindsets
- Encourage staff and raise awareness on the critical importance of diversity of thinking styles, experience, perspectives and expertise
- Develop an idea management system that captures ideas— suggestion box, quality circles

Additional elements to consider ...

- **Flexibility** – to adapt according to the context, culture and sensitivity of those involved;
- **On-going development of the initiative**, if possible by means of self-reflection and self-evaluation and, if possible, action research to evaluate, constructively criticise and modify strategies in order to continually improve them;
- **Motivation on the part of the key people** involved;
- **The establishment of a creative climate** where mutual trust and self-confidence are developed. --“Building a **creative climate** involves systematic development of organisational structures, communication policies and procedures, reward and recognition systems, training policy, accounting and measurement systems and deployment of strategy.” (Tidd, Bessant and Pavitt, Managing Innovation)

Entrepreneurship is closely linked to creativity and innovation ...

- encouraging and developing the entrepreneurial spirit of staff
- the entrepreneurial spirit can be nurtured and developed through the use of simple and effective methods such as the Lateral Thinking techniques
- most people will benefit by adopting a basic set of entrepreneurial practices in their working lives and, where possible, also applying them to the social, civic and commercial institutions within which they are active

Ideas are the currency of success. They separate you from your competition. Edward de Bono

- Techniques of idea generation in group: among the stakeholders of the issue
- Overcoming limited information
- Active learning process
- Countering personal bias
- Acceptance of the solution
- Techniques of idea generation: **Slipcard writing, Brainstorming, Alternative Growth and Stretch matrix, How-to-sessions, Successive filtration of ideas, Nominal group, Delphi technique, GD support systems**

- **Slipcard writing:** Each member of the group receives a pad containing 25 slips or cards. The facilitator can ask the members to write as many alternatives as possible within five minutes. Each alternative must be written on a separate slip or card. Keep the 'how to...' problem as specific as possible. Before and during slip writing, keep hurrying group members and telling them to write as many alternatives as coming to their minds. Instruct "Do not stop to think; you are already thinking. Write whatever comes into your head".
- **Brainstorming:** Four rules govern the brainstorming process. First, all criticisms are ruled out. No one is allowed to judge or question any idea until the idea generation process is completed. Second, 'freewheeling' is welcomed. The emphasis is on creativity and imagination, the wilder the ideas, the better. Third, quantity of alternative solutions is encouraged without reference to quality. The more the alternatives are generated, the greater the likelihood of useful alternatives. Fourth, 'piggy-backing' is good. Everyone is encouraged to suggest how others' ideas can be turned into new ideas and how two or more ideas can be combined to form a new idea.
- **Alternative Growth and Stretch Matrix:** The problem can be further elaborated and stretched for alternatives by tagging different quantitative and qualitative questions (see Table 1). Then the facilitator asks the members in the group to answer these questions.
- **How to session :** Each participant is asked: "Write down on a piece of paper all the different ways you see this problem, also beginning with the words, how to...". The alternative solutions are recorded without censoring.
- **Successive Filtration of Ideas:** Successive filtration of ideas is a funnel-type of idea-selection procedure till the important messages are discerned to sort out an issue or implement a project with procedural details. It is a participant-centred approach.

- **Face-to-face Interacting Group:** In face-to-face interacting group, discussants retrieve ideas, images and thoughts in the form of information from memory, to evaluate the alternative solutions to an issue.
- **Nominal Group:** First, the problem is presented verbally (such as narrating the issue and then posing the question: how can the overtime be reduced and regulated?), pictorially, or written on a flip chart or a blackboard. Second, before any discussion takes place, each member is asked to write an idea/solution independently on the paper for the issue in a fixed time. Third, when all members complete the writing silently, each member takes his/her turn, around the table, presenting a single idea until ideas of all members in the group have been presented and recorded (typically on a flip chart or chalkboard). No discussion takes place until all ideas have been recorded. Each group member is asked to rank the ideas independently according to their importance (from most to least important), assess each idea's usefulness (useful = 2, neutral = 1, not useful = 0), and ease of implementation (easy to implant = 2, somehow easy = 1, not at all easy to implement = 0).

- **Delphi Technique** : First, the problem is identified. A panel of experts is identified on the basis of their knowledge about and experience on the issue. Second, a questionnaire is carefully prepared in the context of the issue/problem to elicit the possible solutions from the experts with the reasons there of. The questionnaire is sent to each expert and each one is asked to provide potential solutions. Each expert anonymously and independently completes the first questionnaire. Third, responses to the questionnaire are sent to the coordinator at a central location. The coordinator transcribes, analyses, and summarises the responses. Four, the coordinator again sends the analysis and summary of responses along with the questionnaire to experts. Five, after viewing the summary, experts are again asked for their responses. It typically triggers new solutions and causes change in the previous solution(s) to the same issue. Lastly, summary of replies along with the questionnaire are sent to experts as often as necessary until a consensual decision/solution emerges from experts.
- **GD Support System**: GD support module is like the nominal group and Delphi technique. Issues are presented to group members through the computer. Group members use personal computers to enter/type responses/replies on computer screen at will, either through simultaneous interaction or over a period of time. The assisted software has many functions such as survey, pooling, voting, and analysis of responses with intuitive and advanced graphics. Depending on requirements of the decision task, the software instantly compile comments, summarise texts, aggregate votes, average ranks, and disseminate solutions. The final responses flash on the screen of all personal computers at the push of the keyboard. **Thank you.**



Critical Thinking and Problems Solving

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Defining Critical Thinking

- Higher order thinking skills like critical thinking and problem solving are considered necessary skills for 21st-century individuals.
- Critical thinking is the intellectually disciplined process of actively and skilfully **conceptualizing, applying, analyzing, synthesizing, and/or evaluating information gathered from, or generated by, observation, experience, reflection, reasoning, or communication, as a guide to belief , judgement and action** (Scriven, 1996)
- Critical thinking is the purposeful, self-regulatory judgment which results in **interpretation, analysis, evaluation and inference**.
- Critical thinking... means making reasoned judgments (Beyer, 1995).

In the book, Critical Thinking, Beyer (1995) elaborately explains what he sees as essential aspects of critical thinking. These are:

- Dispositions: Critical thinkers are sceptical, open-minded, value fair-mindedness, respect evidence and reasoning, respect clarity and precision, look at different points of view.
- Criteria: Apply criteria. Need to have conditions that must be met for something to be judged as believable.
- Argument: Is a statement or proposition with a supporting evidence. Critical thinking involves identifying, evaluating, and constructing arguments.
- Reasoning: The ability to infer a conclusion from one or multiple premises. To do so requires examining logical relationships among statements or data.
- Point of View: The way one views the world, which shapes one's construction of meaning. In a search for understanding, critical thinkers view phenomena from many different points of view.
- Procedures for Applying Criteria: Other types of thinking use a general procedure. Critical thinking makes use of many procedures. These procedures include asking questions, making judgments, and identifying assumptions.

Compared to rote memory tasks, critical thinking takes time.

- Several studies have established that the more proficient one is in using critical thinking the greater one's ability to understand complex phenomena, make important decisions, solve complex problems and resolve difficult issues (Facione et al, 1996, 1998; Paul, 1990, 1992; Cance, 1986).
- Studies have shown that exercising critical thinking fosters a higher level of reasoning and comprehension (Weistein, 1995).

- Critical thinking involves critical implication and discussion, which has a crucial role in activating problem solving and decision making processes (Chaffee, 1994). It goes with logic.
- Critical thinking is a constructivist analysis process to examine what is going on in our environments. This analysis system can be used to define problems, take actions towards an aim, make decisions and conduct retrospective evaluations.
- **Critical thinking problem**
Compare the website of MIT Massachusetts with IIT Kharagpur (gather information, analyze Format, Content, and Usability, make a judgment)

Process of Critical thinking



How to improve critical thinking?

- Seek out ideas- old and new
 - Read books, periodicals, articles, etc. , study innovation, change, and creativity
 - Experiment, brainstorm, have discussions, attend exhibits.
- Take action
 - Dare to be different, challenge assumptions
- Be open minded and flexible
- Apply ideas to every facet of your life
 - Always ask, “How can this be done better?”
- Be curious and observant
 - Look at situation through the eyes of a child
 - Keep asking, “why?”

- Ask good questions
 - Constantly reflect on each aspect
 - Who, what, where, when, why, how, if, etc.
- Develop your reflective thinking skills
 - Daydream about the situation
 - Jump between logical, imaginative, and wild thinking
- Build your knowledge and intuition base
 - Learn how to research and visualize
- Use triggers
 - Find the time and place that help you think best

Problem solving

- Critical thinking is an intentional and reflective way of looking at things or circumstances, while problem solving focuses on a specific situation.
- In every sphere of our life we always face problems those need to be solved.

A problem is:

- a matter about which it is difficult to decide what to do.
- a difficult situation, matter, or person.
- any question or matter involving doubt, uncertainty, or difficulty.
- a question to be answered or solved.

Problem solving is a mental process that involved discovering, analyzing, and solving problems to overcome obstacles and find a solution that best resolves the issue.

The Steps to Problem Solving

- 1- Defining the problem: how to reduce absenteeism in in industry x?
- 2- Gathering information: Multiple
- 3- Studying and analyzing the collected information
- 4- Suggest solutions
- 5- Evaluating the solutions
- 6- Implement solution
7. Evaluate the workability of the solution

- Group discussions, case studies, simulations, structured exercises, critical incidents, controversial debates are some of the tools that can be used in the class rooms to foster critical thinking and problem solving.
- Critical thinkers and problems solvers are not the receivers of information, but the users of information.

Brain mechanisms

- **The problem-solving techniques are most powerful when combined to activate both the logical/rational and intuitive/creative parts of the brain** (Wonder & Donovan, 1984).
- Prefrontal Cortex is also involved in sophisticated **interpersonal thinking skills and the competence required for emotional well-being**. In general, both the left and right sides of the prefrontal cortex are equally involved in social and interactive proficiency.
- Lower back portion of the frontal lobe on each side of the brain, **facilitates creative thinking**.
- The temporal lobe is involved in **reasoning skills, particularly the elaborate task of reading**.
- **Skills for algebraic mathematical tasks and calculations are more prevalent in the left parietal lobe**, while skills for geometric perception and manipulation of three-dimensional figures are **more prevalent in the right parietal lobe**.
- Limbic system is involved in **emotional memory and mood control**.
- THANK YOU